

1.1 What are the marginal distributions of X and Y ?

$$P(X=1) = P(X=1, Y=0) + P(X=1, Y=1) + P(X=1, Y=2)$$

$$= \frac{1}{18} + \frac{1}{12} + \frac{1}{15}$$

$$= \frac{31}{180}$$

$$P(X=1) = \frac{31}{180}$$

$$P(X=2) = \frac{31}{180}$$

$$P(X=2) = P(X=2, Y=0) + P(X=2, Y=1) + P(X=2, Y=2)$$

$$= \frac{1}{18} + \frac{1}{12} + \frac{1}{30}$$

$$= \frac{31}{180}$$

$$P(X=3) = \frac{15}{45}$$

$$P(X=4) = \frac{14}{45}$$

$$P(X=3) = P(X=3, Y=0) + P(X=3, Y=1) + P(X=3, Y=2)$$

$$= \frac{1}{9} + \frac{1}{6} + \frac{1}{30}$$

$$= \frac{14}{45}$$

$$P(Y=0) = \frac{1}{3}$$

$$P(Y=1) = \frac{2}{5}$$

$$P(Y=2) = \frac{4}{15}$$

$$P(X=4) = P(X=4, Y=0) + P(X=4, Y=1) + P(X=4, Y=2)$$

$$= \frac{1}{9} + \frac{1}{15} + \frac{2}{15} = \frac{14}{45}$$

$$P(Y=0) = P(Y=0, X=1) + P(Y=0, X=2) + P(Y=0, X=3) + P(Y=0, X=4)$$

$$= \frac{1}{18} + \frac{1}{18} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}$$

$$P(Y=1) = P(Y=1, X=1) + P(Y=1, X=2) + P(Y=1, X=3) + P(Y=1, X=4)$$

$$= \frac{1}{12} + \frac{1}{12} + \frac{1}{6} + \frac{1}{15} = \frac{2}{5}$$

$$P(Y=2) = P(Y=2, X=1) + P(Y=2, X=2) + P(Y=2, X=3) + P(Y=2, X=4)$$

$$= \frac{1}{15} + \frac{1}{30} + \frac{1}{30} + \frac{2}{15} = \frac{4}{15}$$

2 Are X and Y independent? Justify your answer.

$$P(X=1, Y=0) = \frac{1}{18} \neq P(X=1) \cdot P(Y=0) = \frac{31}{540}$$

X and Y are not independent.

1.3. What is the conditional distribution of X , given that $Y=2$?

what is $E[X|Y=2]$?

$$P(X=1|Y=2) = \frac{P(X=1, Y=2)}{P(Y=2)} = \frac{1}{15} \times \frac{15}{4} = \frac{1}{4}$$

$$P(X=2|Y=2) = \frac{P(X=2, Y=2)}{P(Y=2)} = \frac{1}{30} \times \frac{15}{4} = \frac{1}{8}$$

$$P(X=3|Y=2) = \frac{P(X=3, Y=2)}{P(Y=2)} = \frac{1}{30} \times \frac{15}{4} = \frac{1}{8}$$

$$P(X=4|Y=2) = \frac{P(X=4, Y=2)}{P(Y=2)} = \frac{2}{15} \times \frac{15}{4} = \frac{1}{2}$$

$$E(X|Y=2) = 1 \cdot P(X=1|Y=2) + 2 \cdot P(X=2|Y=2) + 3 \cdot P(X=3|Y=2) + 4 \cdot P(X=4|Y=2)$$

$$= 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{8} + 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{2}$$

$$= \frac{1}{4} + \frac{1}{4} + \frac{3}{8} + 2$$

$$= \frac{23}{8}$$

1.4. Calculate $E(Y)$ and $E(XY)$.

$$E(Y) = 0 \cdot P(Y=0) + 1 \cdot P(Y=1) + 2 \cdot P(Y=2) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{5} + 2 \cdot \frac{4}{15} = \frac{14}{15}$$

$$E(XY) = 0 \cdot \frac{1}{12} + 1 \times \frac{1}{12} + 2 \times \frac{1}{12} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{15} + 2 \cdot \frac{1}{15} + 4 \cdot \frac{1}{30} + 6 \cdot \frac{1}{30} + 8 \cdot \frac{2}{15}$$

$$= \frac{1}{12} + \frac{1}{6} + \frac{1}{2} + \frac{4}{15} + \frac{2}{15} + \frac{2}{15} + \frac{3}{15} + \frac{16}{15}$$

$$= \frac{51}{20}$$

2. 1.4

assume the probability of head on the first toss is P

$$P(A) = P \quad P(B) = P \quad P(C) = P^3 + (1-P)^3 \quad P(D) = 3P^2(1-P)$$

1. A and B

$$P(A) \cdot P(B) = P^2 = P(AB) = P^2 \quad \text{A and B are independent}$$

2. A and C

$$P(A) \cdot P(C) = P \cdot [P^3 + (1-P)^3]$$

$$P(A \cdot C) = P^3$$

A and C are not independent

$$P(A) \cdot P(C) \neq P(A \cdot C)$$

3. A and D

$$P(A) \cdot P(D) = 3P^3(1-P)$$

$$P(A \cdot D) = 2P^2(1-P)$$

A and D are not independent

$$P(A) \cdot P(D) \neq P(A \cdot D)$$

4. C and D

~~$$P(C) \cdot P(D) =$$~~

$$P(C \cap D) = 0$$

C and D are independent